M.M. Golam Hafiz

AI/ML Researcher · Medical Imaging · Federated Learning · Trustworthy AI

Dhaka, Bangladesh · github.com/rafa404 · linkedin.com/in/hafizrafi · Google Scholar

Experience

Research, teaching, and professional roles.

Feb 2026 — present

AI/ML Research Intern · AMIR Lab, American International University-Bangladesh
Research on trustworthy medical AI, personalized federated learning, uncertainty-aware verification, domain generalization, and diffusion modelling under Prof. Dr. Muhammad Firoz Mridha.

Mar 2026 — Jun 2026

Full Stack Web Developer · Agni Systems PLC
Built and maintained production website features using PHP, HTML, CSS, JavaScript, and Bootstrap.

Sep 2024 — Jan 2025

AI Trainer / AI Response Evaluator · Outlier AI
Evaluated model responses for correctness, completeness, groundedness, reasoning quality, and instruction following.

Education

Academic background and qualifications.

2022 — 2026

B.Sc. in Computer Science and Engineering
American International University-Bangladesh · CGPA 3.50 / 4.00

2020

Higher Secondary Certificate, Science
Hajera Taju Degree College · GPA 5.00 / 5.00

Technical Expertise

Tools and methods used across research, modelling, and evaluation.

Programming & frameworks	Python, PyTorch, TensorFlow, Keras, Hugging Face Transformers
Medical imaging	Image classification and segmentation, U-Net architectures, Vision Transformers, transfer learning
Federated & graph learning	FedAvg, personalized FL, GraphSAGE, topology-aware aggregation, prototype-guided personalization
Evaluation	Ablation studies, cross-dataset evaluation, paired statistical testing, calibration, explainability
Generative & diffusion modelling	Diffusion models, conditional diffusion, flow matching, synthetic tabular data generation, constraint-guided generation

Publications

Journal articles, conference papers, preprints, and work currently in peer review.

01 Published Array 2026

Swin-MSEG-Pro: Uncertainty-aware boundary-focused swin transformer with deep supervision for reliable gastrointestinal polyp segmentation
Muhammad Muhtasim Shahriar, **M.M. Golam Hafiz**, M.F. Mridha, Md. Rakibul Islam

02 Published 28th International Conference on Pattern Recognition (ICPR 2026), Main Track 2026

Federated Deep Learning with Client Communication Graphs for Osteosarcoma Histopathology Image Classification
M. M. Golam Hafiz, Muhammad Muhtasim Shahriar, Md. Kishor Morol, Sadek Al Prince, Safia Akter Simi, Dip Nandi, Md. Abdullah Al Jubair

03 **Published** Intelligence-Based Medicine 2026

FedPAXA-Net: Prototype-aligned personalized federated learning for multi-client gastrointestinal endoscopy image classification

Muhammad Muhtasim Shahriar, **M. M. Golam Hafiz**, Md. Rakibul Islam, M. F. Mridha

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04 **Published** 8th International Conference on New Media Studies (CONMEDIA 2025) 2025

Comparative Analysis of Deep Learning Architectures for Brain Tumor Classification Using MRI Images

Muhammad Muhtasim Shahriar, **M. M. Golam Hafiz**, Riadul Islam Rabbi, Tze Hui Liew, Fatema Mostafa Tarin, Ekramul Haque Tusher

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05 **In press** ECCT 2026 2026

End-to-End Deep Learning Framework for Kidney MRI Segmentation and Explainable CKD Detection Using Transfer Learning

Md. Owafeeuzzaman Patwary, Apurbo Biswas, Muhammad Muhtasim Shahriar, **M. M. Golam Hafiz**, Morzia Momo, Kazi Tanvir, Md. Reazul Islam, Dipta Gomes

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06 **In press** International Conference on Computing Advancements (ICCA 2026) 2026

GesturePolypSeg: An HCI Interface for Explainable and Uncertainty-Aware Binary Polyp Segmentation

M. M. Hafiz, Md. Ashikur Rahman Shakib, Md. Maruf Hossain, Muntasir Mehrab, Md. Arafat Rahaman Sifat, Abdus Salam

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07 **In press** IEEE BECITHCON 2026 2026

Comparative Evaluation of CNN, Hybrid, and Transformer Architectures for Nuclei Segmentation in Histopathology

A. K. M. Shahriar Emtiaz, **M. M. Golam Hafiz**, Mamun Mahmud, Muhammad Muhtasim Shahriar, Md. Kishor Morol

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08 **In press** Digital Image Computing: Techniques and Applications (DICTA 2026) 2026

Privacy-Aware Federated Benchmarking and Explainability for Multi-Class Histopathology Image Classification

Muhammad Muhtasim Shahriar, Mamun Mahmud, **M. M. Golam Hafiz**, Dipta Justin Gomes, Md. Mohammad Ali Moni

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09 **Under review** Machine Learning with Applications

DiffBiz-Guard: Inference-Time Numerical Business-Rule Guidance for Auditable Synthetic Tabular Data Release

M.M. Golam Hafiz, Safia Akter Simi, Md. Rakibul Islam, M. F. Mridha

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10 **Under review** Computers & Education: Artificial Intelligence

Reliable Prerequisite-Aware Learning Path Recommendation under Uncertain Knowledge Structures: Long-Horizon Evaluation and Cross-Dataset Validation

M.M. Golam Hafiz, A. K. M. Shahriar Emtiaz, Safia Akter Simi, Md. Rakibul Islam, M. F. Mridha

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11 **Under review** Results in Engineering

DiffusionRef-IMU: Guide-Anchored Residual Diffusion for Cross-Location Inertial Measurement Unit Translation

M.M. Golam Hafiz, Safia Akter Simi, Md. Rakibul Islam, M. F. Mridha

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12 **Preprint** arXiv 2026

FedHisto-PAST: Parameter-Efficient Stain-Aware Federated Learning for Cross-Site Lung Histopathology Classification

Muhammad Muhtasim Shahriar, **M. M. Golam Hafiz**, Saad Aloteibi, Mohammad Ali Moni

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